



Switchable reflective color filters using phase change materials

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Abstract: Phase change materials (PCMs) provide a versatile platform for tunable reflective color filters due to their strong optical contrast between their various states. In this work, we analyze reflective color filter designs based on metal–insulator–metal (MIM) and Fano-resonant cavities using $\text{Ge}_2\text{Sb}_2\text{Te}_5$, $\text{Ge}_2\text{Sb}_2\text{Se}_4\text{Te}_1$ and Sb_2Se_3 . RGB reflective filters containing thin PCM layers were designed and optimized to approximate target color gamut values relative to the sRGB standard. A few selected designs based on MIM cavities and Fano-resonant cavities were fabricated. The measured reflection spectra of the fabricated multilayer structures are in strong agreement with the theoretically predicted reflection spectra.

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1. Introduction

The quest for tunable color filters arises from the increasing demand for versatile and efficient solutions in various applications, such as displays, imaging systems and sensors. Conventional color filters typically rely on static materials with fixed spectral responses, requiring multiple components to achieve the desired color discrimination. In contrast, phase change materials (PCMs) offer a promising alternative by exhibiting reversible transitions between distinct optical states in response to external stimuli, including changes in temperature, electric fields, or light exposure. By incorporating PCMs into micro or nano-structures and by precisely engineering their interactions with incident light, it has become possible to achieve unprecedented levels of control over the transmitted and reflected color and spectral properties. This innovation paves the way for energy-efficient, compact, and dynamically adaptable color filtering solutions that could revolutionize a wide array of industries, such as display technologies and smart eyewear. Despite these favorable characteristics, integrating PCMs into conventional thin film design approaches can pose challenges to how we design color filters due largely to having to deal with two distinct refractive index states and dissimilar extinction coefficients. Hence, this work aims to explore the most effective design methods and structures for leveraging PCMs in color modulation applications. Color depth modulation using $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST-225) and AgIST phase change materials has been previously discussed in the literature. In [1], Rios et al. demonstrated a reflective design comprising a thick metal layer, an ITO spacer followed by a thin PCM layer and an ITO capping layer. These authors demonstrated different colors by changing the thickness of the ITO spacer. Additionally, they demonstrated a change in reflected color when the PCM layer was switched from its amorphous to crystalline state.

A few years later, Jafari et al. [2] presented an RGB reflective filter using GeTe as the PCM. Different structures exhibited different reflective colors after GeTe changed its phase from amorphous to crystalline. They also implemented another reflective color filter design that consisted of two PCM layers (GeTe) separated by an ITO layer with a few additional thin films [3]. They showed that additional colors can be achieved as a result of different index combinations as each PCM layer undergoes phase transition. Subsequently, tunable color filters using Fano resonant cavities were described by Elkabbash et al. [4]. The authors introduced a novel Fano

resonant thin film cavity type as a platform for tunable color filters. Sreekanth et al. [5] proposed that an active thin film nano-cavity induces the Fano resonance, arising from the interaction between a broadband and a narrow-band absorber composed of phase change materials. They demonstrated dynamic color generation using electrically adjustable thin film optical coatings incorporating two distinct PCMs. They further demonstrated the ability to electrically fine-tune this Fano resonance by altering the structural phase of the PCM layers, thereby showcasing active color filters that encompass the entirety of the visible spectrum. Similarly, other studies have also been done by incorporating phase change materials with Fano resonance cavities [6,7].

Beyond conventional display oriented color filtering, PCM based color and spectral modulation has also been extended to broader visible-to-infrared multi functional platforms. A PCM-based multilayer structure has been demonstrated that simultaneously achieves visible color camouflage, solar absorption, and infrared camouflage by exploiting the index contrast between the PCM's amorphous and crystalline states [8]. Along similar lines, a tunable color variable solar absorber based on the chalcogenide PCM Sb_2Se_3 has been reported [9], in which the reflected color and absorption band are modulated through the phase transition. Extending the same design philosophy into multi-band camouflage, a PCM-loaded metamaterial perfect absorber has been introduced [10]. A multi-functional nanostructure incorporating a PCM layer for tunable infrared detection, radiative cooling, and infrared-laser-compatible camouflage has also been demonstrated [11]. Collectively, these works reinforce that PCM-driven color and spectral tuning is a versatile design paradigm spanning visible color filtering, solar absorption, and infrared signature management.

More recent efforts have pushed PCM-based visible–infrared platforms toward adaptive and emissivity tunable operation. A $\text{ZnS}/\text{Ge}/\text{Ag}/\text{Ge}_2\text{Sb}_2\text{Te}_5/\text{Ag}$ multilayer emitter has been proposed [12] that combines visible–infrared compatible camouflage with radiative heat dissipation, where the ZnS thickness sets the reflected color and the GST crystallization fraction independently tunes mid- and long-wave infrared emissivity in the two atmospheric windows. Building on this direction, a dual-PCM (GST/VO_2) multilayer device driven by combined ultrafast-laser and electrical excitation has been demonstrated [13], achieving continuous emissivity tuning from 0.14 to 0.98 and neural-network-driven adaptive infrared thermal camouflage that dynamically matches background signatures. Together with the works discussed above, these studies highlight a clear trend toward PCM-based platforms that unify static visible color filtering with dynamically reconfigurable infrared and thermal signature control.

Filter designs, particularly those incorporating PCMs, typically rely on numerical optimization algorithms, and a structured design methodology has not yet been widely established. This is because most conventional thin-film design approaches consider one optical state at a time, whereas PCM-based filters require simultaneous consideration of at least two optical states. A systematic and physically insightful methodology is therefore important for understanding design constraints and for standardizing the design of reflective tunable color filters, analogous to established approaches for static dielectric filters. Such understanding can improve the efficiency of optimization methods and may also lead to new design strategies. Although the optimal design approach depends on the specific application requirements, switching capability, fabrication constraints, and optical contrast must also be considered when formulating the thin-film structure. In this paper, we outline several design structures, present experimental results, and discuss their associated limitations.

2. Method

PCMs are well suited for reflective filter designs, especially in the visible spectrum, owing to their relatively high absorption in the amorphous state. This same property, however, makes them less suitable for transmission based devices. Strong absorption limits optical penetration depth and disrupts coherent interference, making PCMs less ideal for transmission-based multilayer

filters that rely on precise phase accumulation. In transmissive designs, achieving a consistent and ultra-thin PCM layer is crucial, whereas reflective designs are more tolerant to layer thickness variations. Thermal management is another concern as PCMs require thermal energy for the phase transition. Reflective designs manage thermal effects better than transmissive designs, where the transmissive side can be utilized for designing the appropriate heating/cooling layers without affecting the optical performance.

Although PCMs are inherently switchable, this study focuses primarily on the differences in optical properties between the amorphous and crystalline states rather than on fully reversible device operation. Fully reversible switching, especially through electrical current, is significantly more challenging and requires careful design of thermal time constants and electrical impedance. While such switching is routinely achieved in phase-change memory and metasurface structures, planar thin-film tunable filters are more challenging because of their larger surface coverage areas. Therefore, in this work, we begin with the as-deposited amorphous phase and induce crystallization on a hot plate for experimental evaluation.

Thin films for the stacks in this work were fabricated using RF magnetron sputtering in an argon plasma. All PCM targets (GST, GSST, and Sb_2Se_3) were of stoichiometric composition. Si substrates were cleaned in acetone, methanol, and isopropyl alcohol, then dried under N_2 prior to deposition. The substrates were mounted on a rotating holder to ensure film uniformity, with the working pressure maintained between 3 and 5 mTorr. Deposition rates were calibrated for each target material, allowing individual layer thicknesses to be precisely controlled by adjusting the deposition time.

We analyze the use of PCMs in various reflective color designs such as Fabry-Perot or Metal-Insulator-Metal (MIM) and Fano-resonant cavities using GST-225, GSST, and Sb_2Se_3 . We discuss these different design approaches to achieve reflective color filters for diverse applications.

The optical response of the multilayer thin-film structures was calculated using the transfer matrix method (TMM), where the wavelength-dependent reflection spectra were computed over the 300–1000 nm spectral range for both the amorphous and crystalline states of the phase-change materials. The calculated reflection spectra were subsequently converted into CIE XYZ tristimulus values and CIE 1931 chromaticity coordinates using standard color-matching functions. The corresponding RGB values used for visual color rendering were generated from the calculated XYZ tristimulus values through standard XYZ-to-sRGB conversion.

Two different optimization approaches were employed in this work depending on the design objective. In the first approach, the multilayer structures were optimized to generate reflected colors close to standard sRGB primary colors. In this case, the layer thicknesses were treated as optimization variables and adjusted to minimize the deviation between the simulated chromaticity coordinates and the target sRGB chromaticity coordinates in the CIE 1931 chromaticity diagram. Following the optimization, the optical responses corresponding to both the amorphous and crystalline states were calculated to evaluate the reflected color shift and optical contrast induced by the PCM phase transition.

In the second optimization approach, the multilayer structures were directly optimized to maximize the reflected color contrast between the amorphous and crystalline PCM states. In this case, the layer thicknesses were swept and optimized to maximize the separation between the reflected colors corresponding to the two PCM phases. The optimization objective function was defined as

$$f = \frac{1}{\|\mathbf{A} - \mathbf{B}\|} \quad (1)$$

where $\mathbf{A} = [X_1, Y_1, Z_1]$ and $\mathbf{B} = [X_2, Y_2, Z_2]$ represent the CIE XYZ tristimulus vectors corresponding to the amorphous and crystalline states, respectively. Minimizing this objective function effectively maximizes the Euclidean distance between the two colors in XYZ color space, thereby increasing the reflected color contrast after the PCM phase transition.

To quantitatively evaluate the perceptual color contrast between the two optical states, the calculated XYZ tristimulus values were additionally converted into the CIELAB color space using the D65 standard illuminant, and the corresponding perceptual color difference was calculated using the CIE76 ΔE color difference metric:

$$\Delta E = \sqrt{(L_1^* - L_2^*)^2 + (a_1^* - a_2^*)^2 + (b_1^* - b_2^*)^2} \quad (2)$$

where (L^*, a^*, b^*) represent the CIELAB color coordinates corresponding to the two optical states of the PCM. Larger ΔE values indicate stronger perceptual color contrast between the reflected colors before and after the phase transition.

All optimizations were performed using MATLAB's `fmincon` constrained nonlinear optimization solver. The multilayer thicknesses were constrained within experimentally realizable lower and upper thickness bounds corresponding to fabrication limitations. The same optimization framework was consistently applied across the different thin-film architectures investigated in this work, including Fabry-Perot, MIM, and Fano-resonant cavity structures.

3. Modeling results and discussion

3.1. RGB filters

Achieving accurate color representation on displays relies on RGB (red, green, blue) filters to create a broad spectrum of colors. The specific requirements for these filters vary based on the application and desired color precision. They must cover a significant portion of the human-perceived color space, adhering to standards like sRGB or Adobe RGB for accuracy.

Therefore, we explored designing RGB filters using commonly used PCMs to approximate the achievable color gamut based on fundamental thin film models. We optimized the layer thicknesses to match specified color gamut values for each channel. However, our optimization process did not consider the phase change property of the PCM. As a result, there are cases where the contrast between two states may not be significant due to the relatively thin PCM layers. This limitation impacts the overall spectral shift and, consequently, the magnitude of the color shift which will be discussed in later section with different examples. However, our main objective in designing these RGB filters is to show the range of colors that can be achieved by each PCM in the color gamut with respect to the sRGB standard.

3.1.1. Metal-insulator-metal structures

Metal-insulator-metal (MIM) cavities offer several advantages in optical filtering and modulation applications. They are characterized by strong optical field confinement within an ultrathin dielectric spacer, which enables pronounced resonant behavior at subwavelength thicknesses. In thin-film implementations, MIM cavities rely on multiple reflections between metal interfaces to form optical resonators, where resonance conditions are primarily determined by the optical path length and refractive index of the insulating layer. By modifying the spacer thickness or its refractive index, precise control over the resonant wavelength can be achieved, making MIM cavities well suited for tunable and switchable devices. Their compact geometry allows for straightforward integration into planar optical systems. Because of their strong sensitivity to changes in the optical properties of the dielectric layer, MIM cavities are also attractive for sensing and modulation applications. Despite operating in a high-loss regime, MIM cavities maintain a relatively simple layered structure, which is advantageous for filter fabrication and large-area integration [14].

MIM structures are a useful choice for reflective color filters in optical and photonic applications. These structures are capable of providing precise and selective color filtering, allowing the reflection of specific colors within the visible spectrum while efficiently suppressing others. Their customizability is a significant advantage, as their design parameters can be fine tuned,

such as insulator layer thickness and metal layer dimensions, to create a diverse range of colors to address specific needs. MIM color filters are compact, making them suitable for integration into various optical devices like displays, cameras, and sensors. Their angle-independent performance ensures consistent color accuracy across different viewing angles, while their low optical losses maximize color vibrancy. These structures can be mass-produced using established nanofabrication techniques, enabling their use in a wide array of applications, including LCDs, image sensors, and optical sensors, where precise and efficient color reproduction is essential.

For the MIM design used in the study, silver (Ag) was selected as the primary metal layer and indium tin oxide (ITO) as the dielectric layer. Despite its high conductivity, ITO was selected as the insulation layer primarily due to its elevated refractive index and minimal extinction coefficient. This choice originated from the potential to utilize its electrical conductivity for electrically switching the PCMs. Nonetheless, PCMs possess a higher refractive index and an extinction coefficient that resembles those of metals, thus used to effectively function as a metal within this structure. Consequently, by implementing a straightforward optimization algorithm, these structures were fine-tuned to attain standard RGB color values within the color gamut. sRGB standards were used as a reference or guideline when designing these filters and the chromaticity coordinates for each primary color for sRGB space are as follows:

The MIM configuration allows the PCM to function as either the top (Fig. 1(a)) or bottom metal layer (Fig. 1(b)), offering two potential setups. Each configuration was explored to determine the most favorable outcome. The optimization process was aimed to attain sRGB values as outlined in Table 1. Figure 2 displays the reflected colors obtained for each PCM phase (amorphous and crystalline) when the PCM served as the top metal layer. These colors were achieved through optimization specifically targeting sRGB colors, without considering the contrast between the two PCM states as an optimization parameter. Consequently, in most cases depicted in Fig. 2, the contrast between the PCM states is not pronounced. Nevertheless, these results emphasize the design flexibility provided by each phase change material in terms of color gamut coverage for the MIM structure, as illustrated in Fig. 3(d). Abdelraouf et al. have published similar research utilizing a metal-insulator-metal (MIM) cavity incorporating a low-optical loss phase change material (PCM), specifically Sb_2S_3 , for RGB color filter applications. This setup enables optical switching between phases, expanding its versatility and potential applications in color filtering and optical modulation [15].

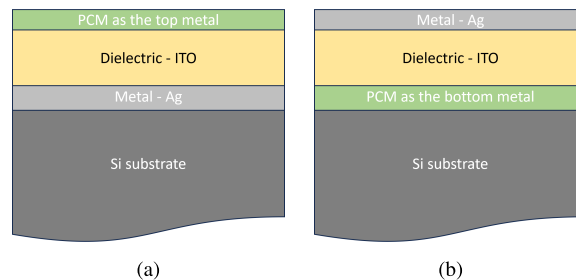


Fig. 1. Two configurations of the MIM cavity structure (a) structure with the PCM as top metal layer; (b) structure with the PCM as bottom metal layer.

Figure 4 depicts the optimized outcomes for the MIM structure utilizing the PCM as the bottom metal layer. The spectral response and the corresponding color gamut coverage is shown in using Fig. 5. Using a PCM as the bottom metal layer offers significant benefits when implementing these filters, as it promotes a more convenient switching mechanism. Additionally, having the PCM layer closer to the Si substrate could be beneficial for fast heat dissipation during the amorphization process (melt-quench reverse switching) using the melt quench technique. As

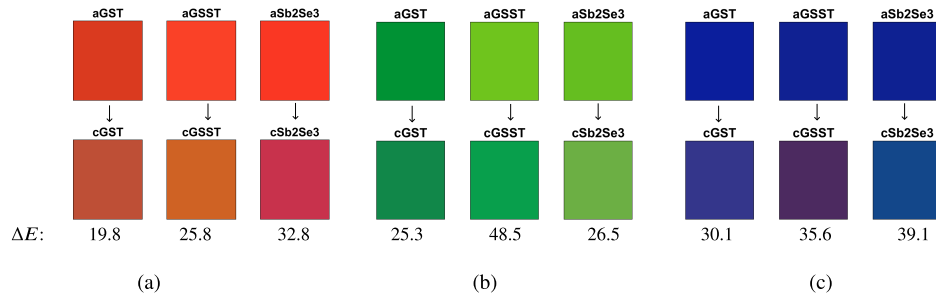


Fig. 2. Reflected colors of MIM filters on a Si substrate with the stack Ag/ITO/PCM, designed using different PCMs and optimized for the sRGB values of (a) red, (b) green, and (c) blue. The layer thickness ranges are Ag(30–146 nm), ITO(128–335 nm), and PCMs: GST (10 nm), GSST (10–12 nm), and Sb₂Se₃ (10 nm).

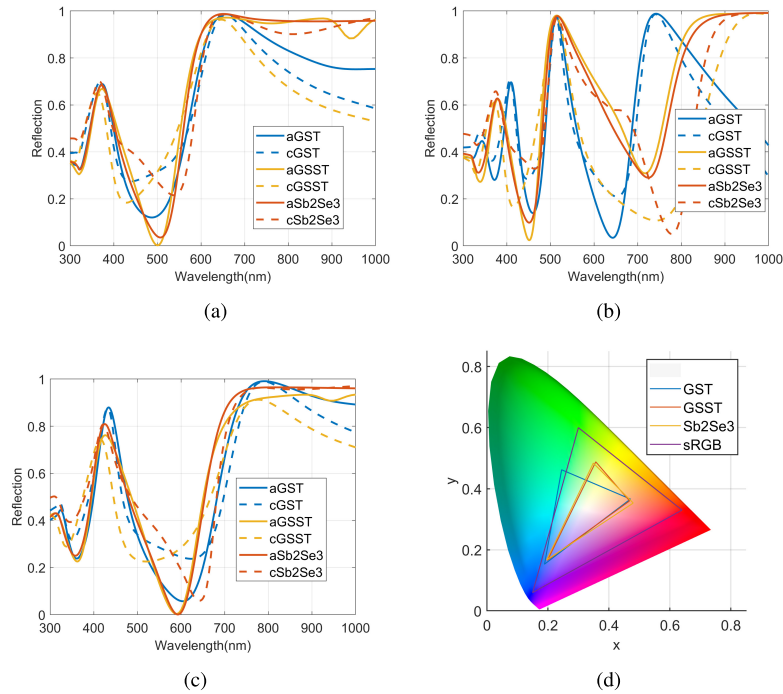


Fig. 3. Reflection spectra of the (a) red, (b) green, and (c) blue MIM filters with PCM as top layer, as shown in Fig. 2. (d) Comparison of the color gamut for each PCM with respect to sRGB.

Table 1. Standard sRGB values for each color

(x_r, y_r)	(0.64, 0.33)
(x_g, y_g)	(0.29, 0.60)
(x_b, y_b)	(0.15, 0.06)

these are reflective filters, introducing a metal heater can be done without compromising the optical response of the filter. In fact, it can potentially enhance the output color in terms of color contrast between two states. These findings will be discussed in subsequent sections of the paper.

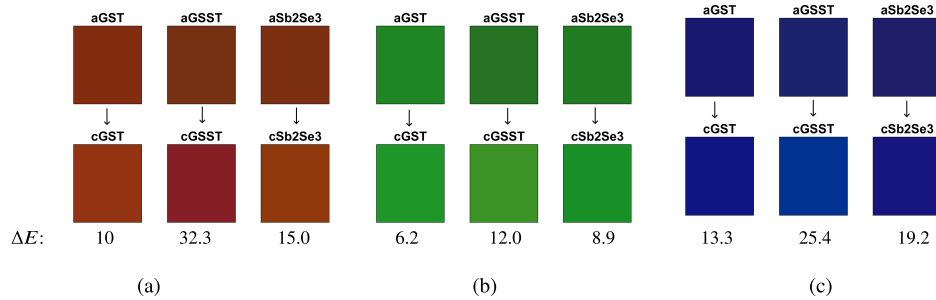


Fig. 4. Reflected colors of the MIM filters on Si substrate with stack PCM/ITO/Ag, designed using different PCMs and optimized for (a) red, (b) green, and (c) blue sRGB values. The thickness ranges are PCMs: GST(93-110 nm), GSST(25-100 nm) and Sb₂Se₃ (91-108 nm), ITO(63-335 nm), Ag(10 nm)

Similarly, incorporating a PCM layer within the cavity enables us to readily alter the resonance condition in the Fabry-Perot cavity as it transitions between its amorphous and crystalline states. By strategically placing the PCM layer within the cavity and adjusting its thickness, we can regulate the extent to which it affects the resonance condition, thereby leveraging it to induce variations in color during phase transitions (Fig. 6). However, systematic design methodologies for Fabry-Perot cavities primarily focus on designing transmission filters, given that reflection is minimized at resonance wavelengths. Consequently, when developing reflection filters, we must depend on optimization techniques to achieve desired performance. The optimized structures for sRGB values using GST, GSST and Sb₂Se₃ are shown in the Fig. 7, while their spectral response and color gamut coverage is shown in Fig. 8. Similarly to the MIM designs discussed above, ITO was selected as the cavity material as its electrical conductivity can be useful to switch PCM layer [1]. While these structures can also be designed using other dielectric materials such as TiO₂ and SiO₂, alternative switching mechanisms such as optical switching would need to be used.

3.1.2. Fano-resonance cavity

Elkabbash et al. introduced a novel thin film cavity type [4], which generates Fano resonance between two coupled cavities. This structure comprises a Fabry-Perot cavity alongside a high-index absorptive thin layer, effectively combining the strong interference effect of a broadband absorber with the narrowband filter [16–19]. When appropriately configured with specific thicknesses and material properties, these two cavities can couple and resonate to generate Fano resonance. Physically, the Fano resonance originates from interference between two spectrally different resonant pathways: a broad continuum-like resonance and a narrow discrete resonance mode. In the present structure, the absorptive metallic cavity formed by the thin Ag layers supports a broadband optical response with relatively low quality factor resonance behavior, while the dielectric cavity region containing the PCM supports a narrower Fabry-Perot-like resonant mode with stronger wavelength selectivity. Optical coupling between these two resonances leads to constructive and destructive interference at nearby wavelengths, producing the characteristic asymmetric Fano line shape observed in the reflection spectrum.

The thin Ag layers play multiple roles within the cavity. In addition to providing broadband reflection and optical confinement, the metallic layers introduce controlled absorption that broadens the continuum resonance response. Meanwhile, the PCM layer contributes strong

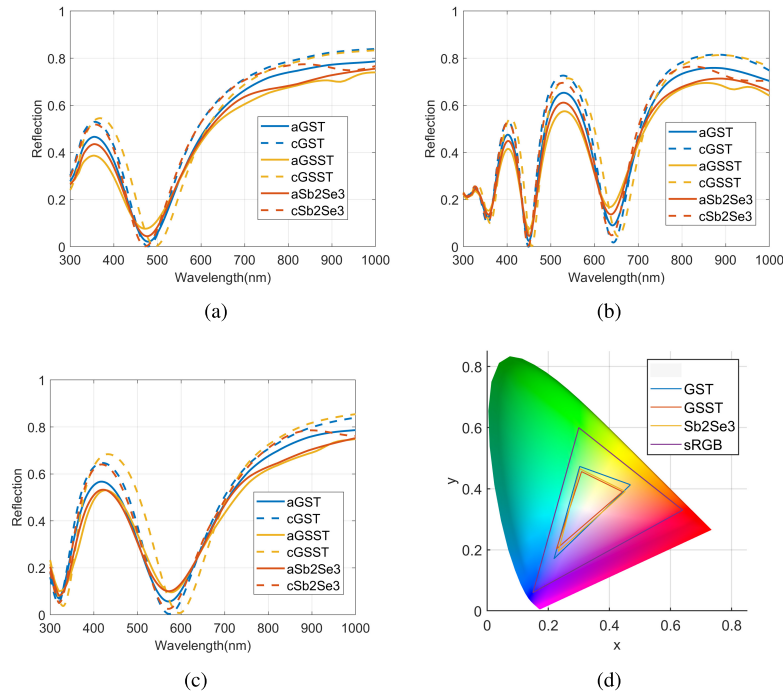


Fig. 5. Reflection spectra of the (a) red, (b) green, and (c) blue MIM filters with stack PCM/ITO/Ag in Fig. 4. (d) Comparison of the color gamut for each PCM with respect to sRGB.

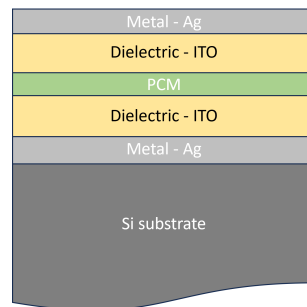


Fig. 6. Thin film structure of a Fabry-Perot resonant cavity with a PCM layer at center.

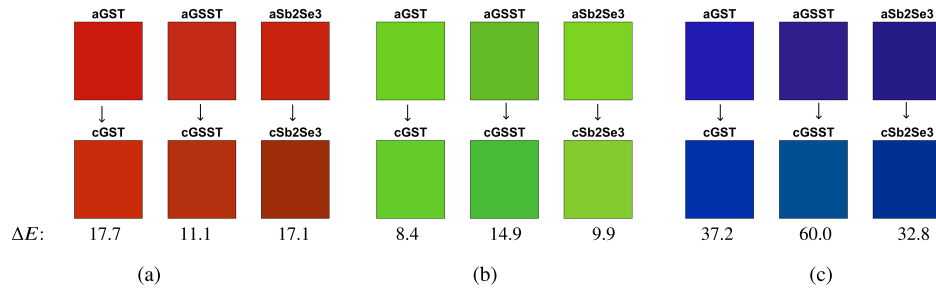


Fig. 7. Reflected colors of the MIM on Si substrate with multilayer stack Ag/ITO/PCM/ITO/Ag using different PCMs optimized for (a) red, (b) green, and (c) blue sRGB values. The thickness ranges are Ag(10-168 nm), ITO(66-163 nm), PCM: GST(10-24 nm), GSST(10-32 nm) and Sb₂Se₃(11-33 nm), ITO(61-145 nm) and Ag(10 nm)

refractive index contrast and phase accumulation within the cavity. During phase transition, the refractive index and extinction coefficient of the PCM change significantly, modifying the optical path length and resonance coupling conditions within the cavity. As a result, both the resonance wavelength and spectral asymmetry can be tuned, leading to strong reflected color modulation between the amorphous and crystalline states.

The intriguing aspect is that this phenomenon results in both reflection and transmission peaks occurring at the same wavelength and capable of producing bright colors [6]. Therefore, in this paper, we investigate the performance of the Fano resonance cavity (Fig. 9) for each PCM when optimizing for the sRGB values. In contrast to other structure types, in the Fano resonance cavity, the reflectance color directly corresponds to the output of the reflection peak, whereas in other structures, the reflectance color results from a combined effect of the reflection spectrum. Thus we have more control over the resonance wavelength. The optimized results for the RGB filters for each PCM are depicted in Fig. 10. The results are further detailed in Fig. 11. When comparing the coverage of the color gamut, the Fano resonance structure can cover a larger area within the gamut compared to other structures discussed.

3.2. Color filters optimized for contrast

In this section, we have re-optimized all the structures previously discussed to maximize the contrast between the amorphous and crystalline states of the PCM. This process involved calculating the reflected color for each state of the PCM and optimizing the structure's thickness until the difference between color values was maximized. Figure 12 shows the color outputs obtained by optimizing the Si substrate/Ag/ITO/PCM structure for color contrast. Higher contrast is influenced by the higher change in the refractive index and extinction coefficient between the amorphous and crystalline state of the PCMs. Additionally, the location of the PCM layer within the cavity and its thickness will also impact the outcome as it determines how much it would interfere with the electric field distribution within the thin film structure. However, for the metal/dielectric/PCM structure, a moderate contrast was observed in the optimized results.

Next, the optimization process was applied for the Si substrate/PCM/ITO/Ag structure. However, resulted color contrasts were mainly from intensity difference of the reflection spectrum between two states rather than reflectance peak shift. Therefore, the structure was modified by adding a Au layer underneath (as it can be used as a metal heater for continuous switching) and re-optimized. The color outputs of Si substrate/Au/PCM/ITO/Ag structure shown in Fig. 13 demonstrate comparatively larger contrast compared to the PCMIM structure. Therefore, addition of Au layer helped in improving contrast in addition to be useful in reverse switching.

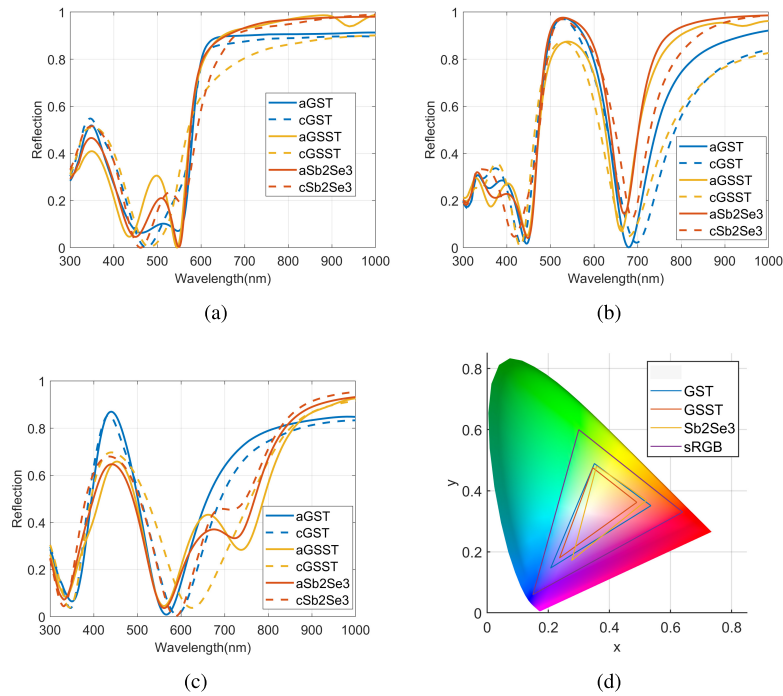


Fig. 8. Reflection spectrum of the (a) red, (b) green, and (c) blue multilayer stack Ag/ITO/PCM/ITO/Ag shown in Fig. 7. (d) Comparison of the color gamut for each PCM with respect to sRGB.

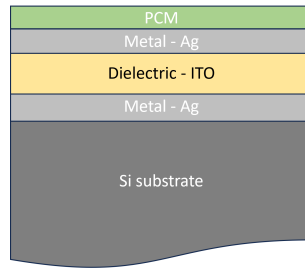


Fig. 9. Fano resonant thin film structure.

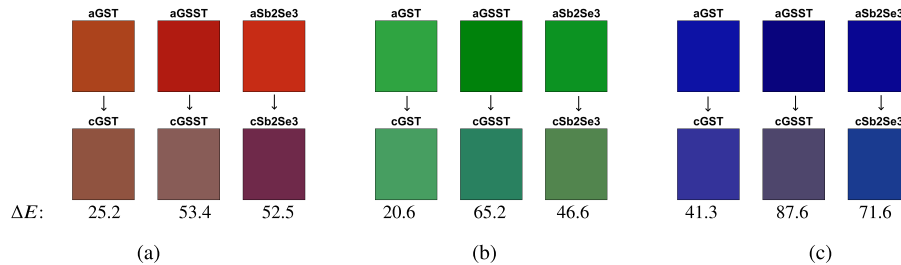


Fig. 10. Reflected colors of Fano filters on a Si substrate with the structure Ag/ITO/Ag/PCM, designed using different phase-change materials (PCMs) and optimized for the sRGB values of (a) red, (b) green, and (c) blue. The layer thickness range are Ag (150 nm), ITO (58-132 nm), Ag (10-26 nm), PCMs: GST (10-12 nm), GSST (15-16 nm) and Sb₂Se₃ (10-12 nm)

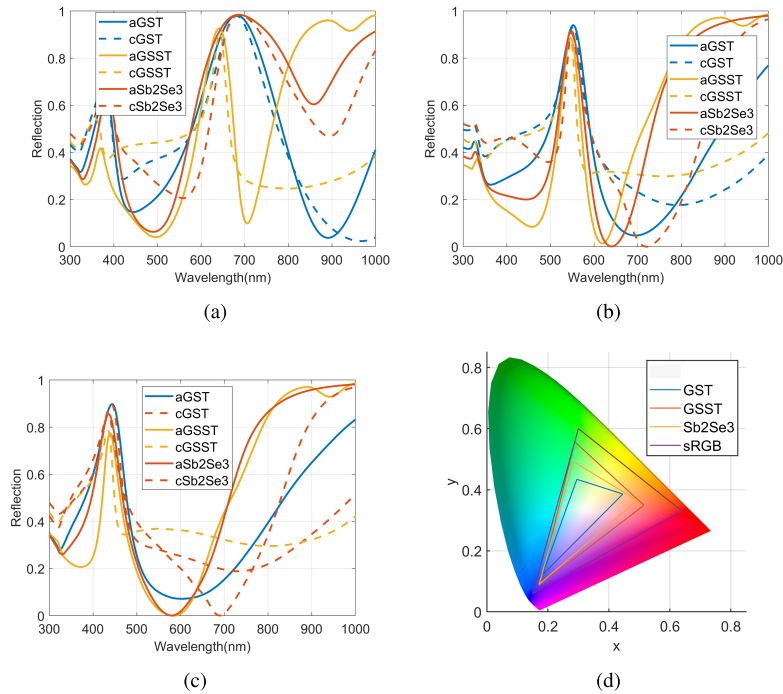


Fig. 11. Reflection spectra of the (a) red; (b) green; and (c) blue Fano filters shown in Fig. 10. (d) Comparison of the color gamut for each PCM with respect to sRGB.

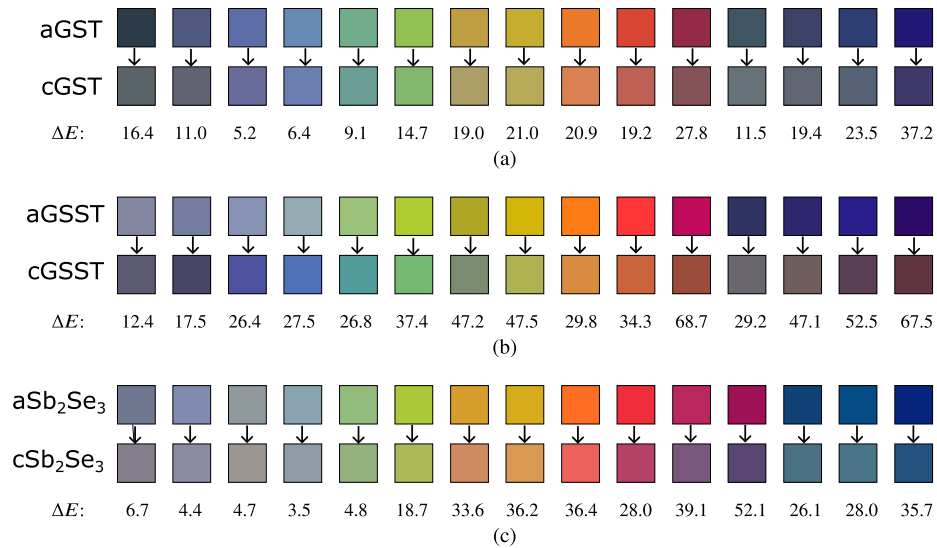


Fig. 12. Color contrast optimization results for Ag/ITO/PCM multilayer designs on a Si substrate, using different PCM materials: (a) GST, (b) GSST, and (c) Sb₂Se₃. The explored thickness ranges are Ag (29–100 nm), ITO (10–160 nm), and PCM: GST/GSST (10–50 nm) and Sb₂Se₃ (11–25 nm). The reported ΔE values correspond to the calculated CIE76 perceptual color difference between the amorphous and crystalline states.

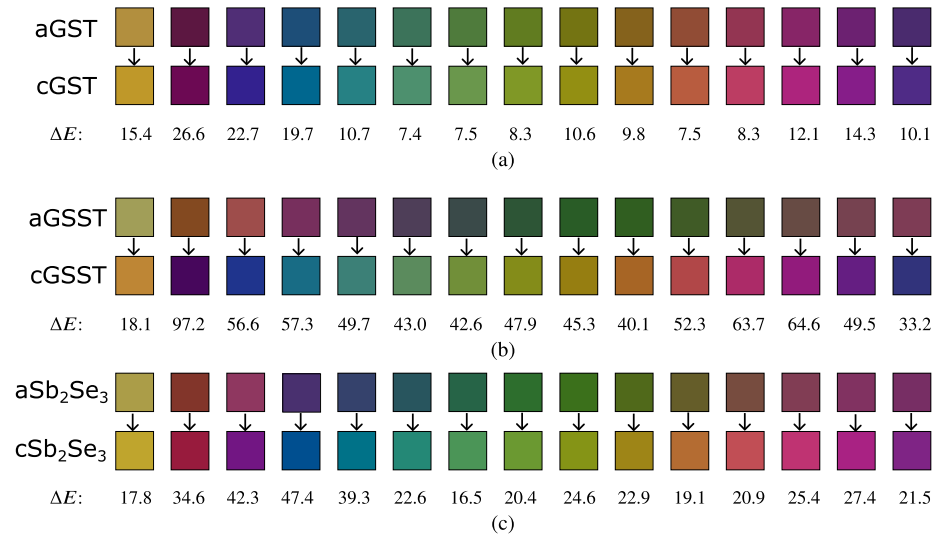


Fig. 13. Color contrast optimization of Au/PCM/ITO/Ag multilayer stacks on a Si substrate for different phase-change materials: (a) GST, (b) GSST, and (c) Sb_2Se_3 . The layer thicknesses are fixed at Au (50 nm) and PCM (25 nm), while ITO and Ag vary within 66–220 nm and 10–22 nm, respectively.

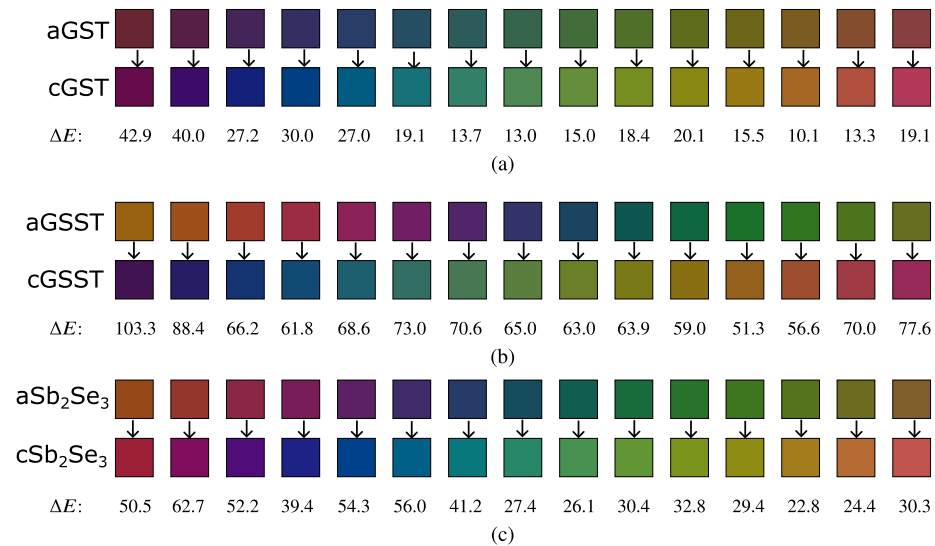


Fig. 14. Optimization of color contrast in Ag/ITO/PCM/ITO/Ag multilayer structures deposited on Si, comparing (a) GST, (b) GSST, and (c) Sb_2Se_3 as the PCM layer. The structure consists of Ag (30 nm) / ITO (22 nm) / PCM (18 nm) / ITO (80–192 nm) / Ag (10 nm), with the upper ITO thickness varied.

Based on previous discussions, it is evident that a cavity design with a PCM layer in the center has more design freedom when it comes to tunable color filters. Therefore, in proving that when this structure was optimized for color contrast, we were able to obtain greater contrasts compared to other structures we've discussed so far. Amongst all the designs, GSST showed the highest contrast and some of these results are depicted in the Fig. 14.

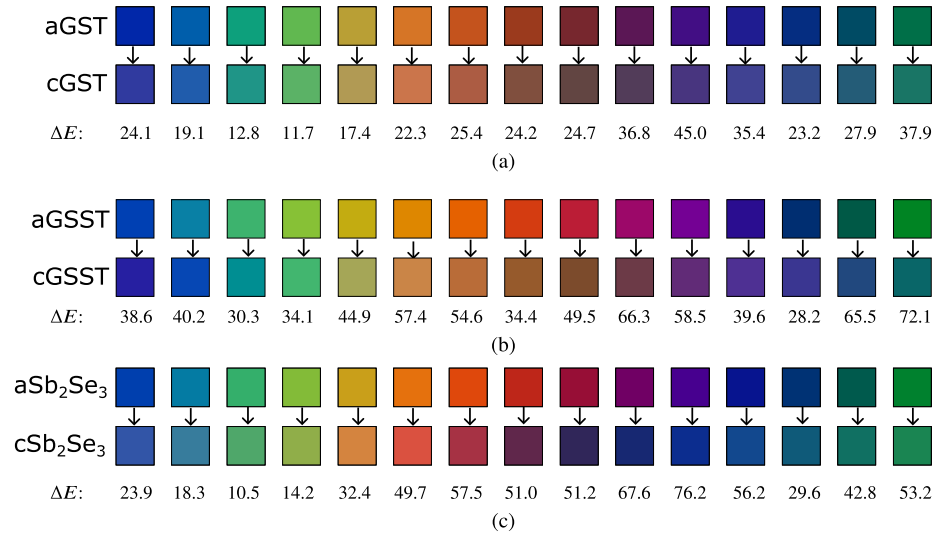


Fig. 15. Color contrast optimization results for Ag/ITO/Ag/PCM multilayer structures on a Si substrate with different PCMs: (a) GST, (b) GSST, and (c) Sb₂Se₃. The thickness ranges are Ag (50–75 nm), ITO (65–205 nm), intermediate Ag (10–12 nm), and PCM: GST/GSST (10–12 nm) and Sb₂Se₃ (10–25 nm).

Similarly, the Fano resonance structure was optimized for color contrast as well (Fig. 15). This design type provided high reflective color designs but the contrast achieved is not significant as in Fabry-Perot cavity with a PCM. This could also be limited by the capacity of the optimization algorithm we used. However all these results with different optimizations indicate that these materials have potential to be used for variety of color applications.

4. Experimental results and discussion

To validate the proposed designs and demonstrate their practical realization, several configurations were implemented using GST and GSST phase-change materials. All designs were fabricated by magnetron sputter deposition in an argon plasma. The PCM layers and indium tin oxide (ITO) were deposited using 3-inch stoichiometric targets obtained from plasmaterials, and the stoichiometry of the materials was verified using energy-dispersive X-ray analysis (EDX).

The GSST, GST, and ITO were deposited under an RF discharge power of 100W at a pressure of 4mT in an Ar⁺ plasma, while a DC power of 25W was applied for the Ag target. Refractive indices were assessed using an ellipsometer, revealing values closely resembling those documented in the literature for both GSST and GST. In the most of designs, Ag layers were relatively thin. To enhance adhesion and achieve uniformity, a thin copper (Cu) layer of approximately 1nm thickness was included before each Ag layer, as suggested by previous studies [20,21]. Copper was chosen for this purpose due to its minimal interference with the optical response compared to alternatives such as germanium (Ge) and chromium (Cr). In the fabrication process, the thickness of each layer was monitored post-deposition using a stylus profilometer to ensure accuracy. Additionally, spectral measurements were conducted using a Filmertics reflection/transmission

thin film analyzer. The overall functionality of the final device was assessed under both amorphous and crystalline states for both GST and GSST. The performance of the final device was evaluated under both amorphous and crystalline states of both GST and GSST. To demonstrate the phase change effect, the films were annealed on a hot plate. In practical applications, the phase change can be accomplished through methods such as electrical joule heating or optical heating. While our study did not directly investigate these alternative switching techniques, the design process involved thoughtful consideration of material selection and certain modifications with the intention of accommodating these switching techniques.

Figures 16 and 17 depict fabricated filter designs chosen from numerous optimized designs. The images demonstrate the feasibility of fabricating these designs and achieving the anticipated color contrast. The measured reflection spectra of these filters were compared with their theoretical response that was generated from the transfer matrix method (TMM), confirming their good agreement.

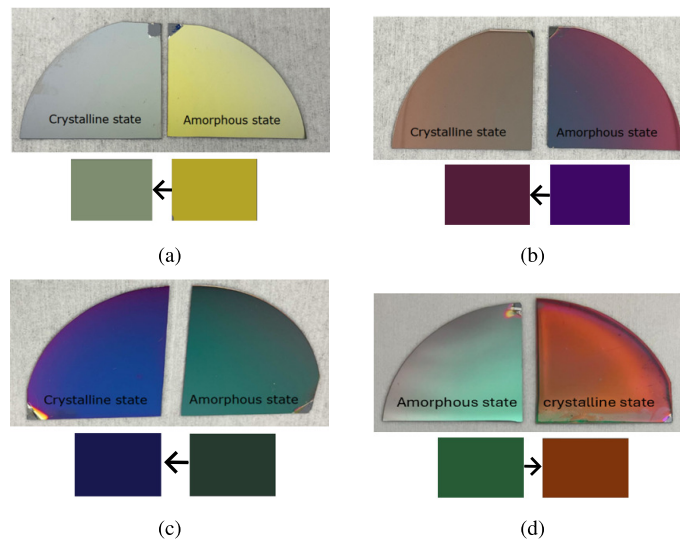


Fig. 16. Comparison between fabricated and modeled reflective color filter responses before and after annealing. All multilayer structures are deposited on Si substrates with a thickness tolerance of ± 5 nm. The layer stacks are defined as: (a) Cu (~ 1 nm) / Ag (125 nm) / ITO (80 nm) / GSST (34 nm); (b) GSST (62 nm) / ITO (77 nm) / Cu (~ 1 nm) / Ag (10 nm); (c) Cu (~ 1 nm) / Ag (145 nm) / ITO (142 nm) / GST (13 nm) / ITO (86 nm) / Cu (~ 1 nm) / Ag (10 nm); and (d) Cu (~ 1 nm) / Ag (33 nm) / ITO (140 nm) / GSST (18 nm) / ITO (56 nm) / Cu (~ 1 nm) / Ag (10 nm).

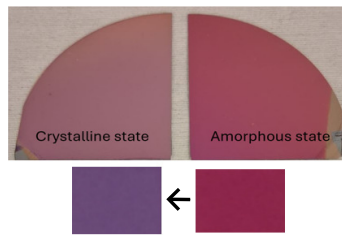


Fig. 17. Fabricated Fano resonant cavity that consist of Si substrate/ Cu (2 nm) / Ag (88 nm) / ITO(185 nm) / Cu (2 nm) / Ag (25 nm) / GSST (10 nm). Thickness tolerance of all layers is ± 5 nm.

5. Conclusions

We have explored the potential of phase change materials in reflective color filter designs due to their refractive index properties and the advantages they offer over different design types. Nevertheless, there is currently no generalized direct design method for creating reflection filters apart from optimization techniques. In this study, we aimed to highlight various optimization criteria applicable in designing tunable reflective color filters, considering both the amorphous and crystalline structural states. Many of the findings we discussed were predominantly constrained by the efficacy of the optimization algorithms employed which were developed and implemented using MATLAB. Nonetheless, the potential for enhancing these design algorithms exists through the integration of machine learning approaches.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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